A More Construction Details

A.1 Data Filtering

To ensure the quality of the generated images, we leverage the advanced image-text alignment metric, VQAScore [28], for data filtering. Specifically, we use two strong foundation models, GPT-4o [35] and CLIP-FlanT5 [28], which achieve high performance comparable to human evaluation as reported by Lin et al. [28], to independently compute alignment scores and then take their average. Images with an average score below the threshold α= 0.85 or with high inter-model variance exceeding β=0.5 are flagged for manual review. For images containing text, we further enhance filtering precision using a professional OCR tool, iFLYTEK OCR [20], to ensure accurate text recognition and alignment.

To assess the effectiveness of our filtering pipeline, we randomly sampled 500 images from the final dataset for manual review. The evaluation criteria included: (1) whether the image content aligns with the corresponding prompt, and (2) whether the image could potentially mislead models to incorrect answers. Our review found that nearly all images demonstrated good alignment and quality. Fewer than 5% exhibited minor issues, such as small distortions or artifacts caused by generation model limitations, which were unrelated to semantic alignment. Considering these minor imperfections are unlikely to affect evaluation outcomes, and in light of known limitations in widely-used datasets, e.g., at least 2,916 labeling errors in the ImageNet validation set [50], we consider the quality of our dataset acceptable for rigorous hallucination evaluation.

A.2 Instruction Templates

We design type-specific instruction templates for four question formats: yes-or-no questions (YNQ), multiple-choice questions (MCQ), free-form questions (FFQ), and image captioning (IC), supporting both discriminative and generative tasks.

Specifically, for YNQ, we construct questions that generate "yes" or "no" answers, such as "*Is the <entity> <attribute> in the image?*", "*Is the <entity1> <relation> <entity2> in the image?*", "*Is the <fact> about <entity> true in the image?*". For MCQ, we randomly shuffle the ground-truth answer with three distractors drawn from the same candidate element set. Example templates include "*What is the <attribute> of the <entity> in the image? <Options>*", and "*What is the <fact> about the <entity> in the image? <Options>*". For FFQ, we simply remove the options from the MCQ templates to form open-ended questions. For IC, the instruction is a straightforward prompt, "*Please describe the image.*"

A.3 Evaluation Details

For discriminative tasks, we adopt accuracy as the evaluation metric by directly extracting and matching the corresponding options from model responses. For generative tasks, we follow the LLM-as-a-Judge protocol in prior works [29, 42], i.e., given the image content, instruction, and ground-truth answer, we prompt GPT-4o [35] to determine whether the model's response is hallucination-free. The non-hallucination rate is then used as the evaluation metric. The detailed evaluation prompt template is illustrated in Figure 4.

###Task:
Given the input instruction, the ground truth answer, and image information, evaluate whether the model response is aligned with the image content.

###Evaluation Criteria:
Without hallucination: The response is semantically similar to the ground truth answer and does not contain any contradictory factual claim with the provided information.
With hallucination: The response is completely different from the ground truth answer, or contains contradictory factual claim about the object, action, attribute, or any other detail that is not grounded in the provided information.
Unclear: The response does not give a specific answer, or refuse to answer the question.

###instruction: [INSTRUCTION]
###Ground Truth Answer: [GROUND TRUTH]
###Image Information: [IMAGE PROMPT]
###Related Facts: [FACTS] (if applicable)
###Model Response: [MODEL RESPONSE]
###Output Format:
With Hallucination/Without Hallucination/Unclear, [Evidence].

Figure 4: Evaluation prompt template for generative task.

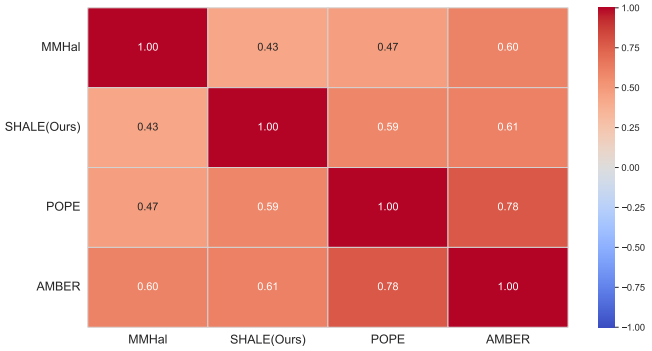

Figure 5: Visualization of the correlation between hallucination benchmark evaluation results across different LVLMs.

B More Experimental Results

B.1 Correlation with Other Benchmarks

To further validate the effectiveness of SHALE, we analyze the correlation between its evaluation results and those from other hallucination benchmarks. Specifically, we compute the Pearson correlation coefficients between SHALE's assessments of faithfulness hallucination and the corresponding results from existing benchmarks on the same set of models. The correlation matrix is visualized in Figure 5.

The Pearson coefficient ranges from [-1,1], where 1 indicates a perfect positive correlation, -1 a perfect negative correlation, and 0 no correlation. Our analysis reveals a strong positive correlation between SHALE and other benchmarks, demonstrating the validity

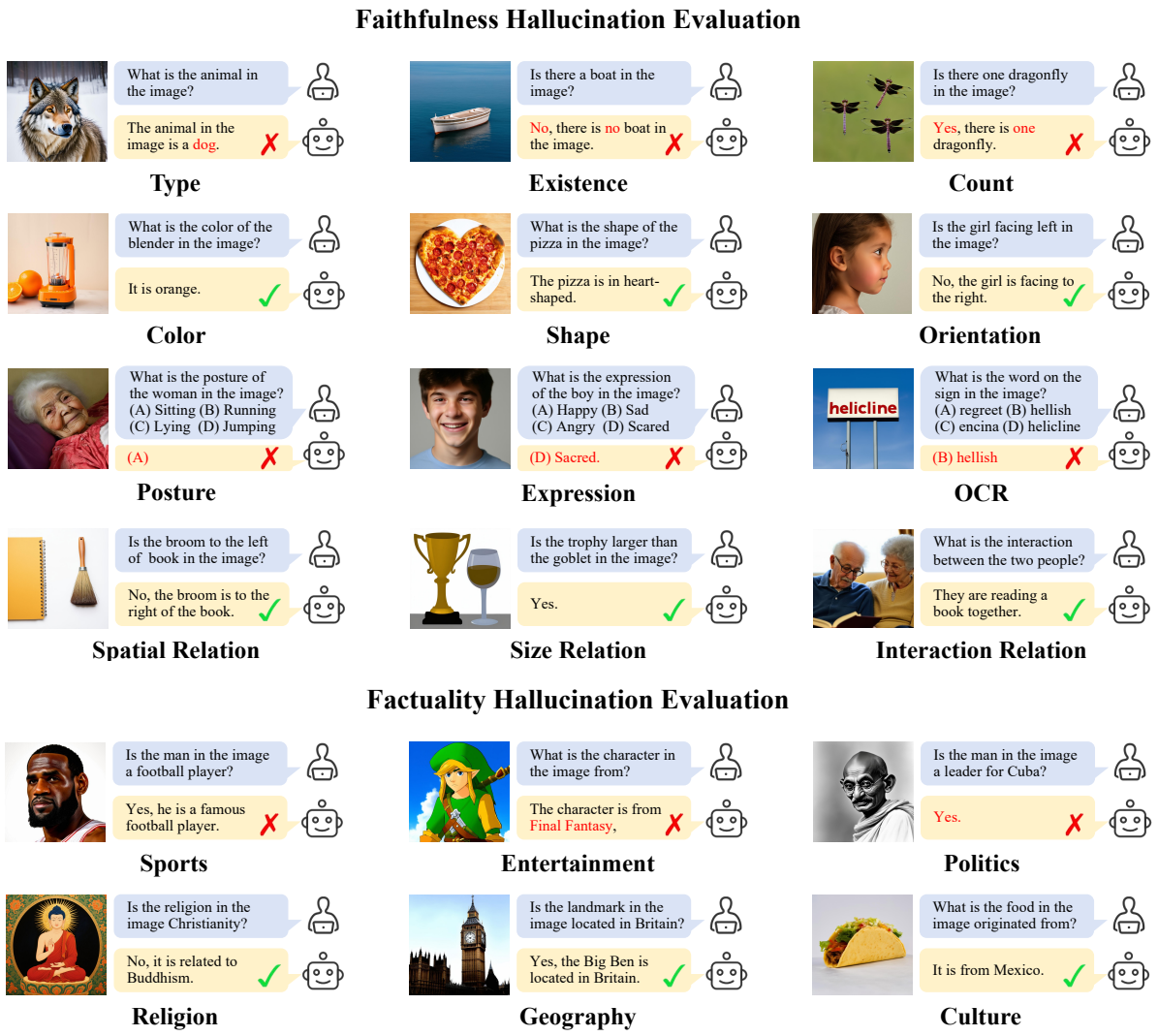

Figure 6: Examples of SHALE evaluation across faithfulness and factuality hallucination.

of using synthetic datasets for hallucination evaluation in LVLMs. Furthermore, compared to these existing benchmarks, SHALE encompasses a broader spectrum of hallucination types, evaluation tasks, and scenarios, providing a more comprehensive evaluation framework.

B.2 More Evaluation Examples

We provide examples of SHALE evaluation across 12 faithful visual perceptual aspects and 6 factual knowledge domains in Figure 6.

B.3 Fine-grained Evaluation Results

We provide detailed evaluation results under clean scenario across specific question formats, fine-grained visual perceptual aspects, and factual knowledge domains in Table 5, Table 6, and Table 7, respectively.

Table 5: Evaluation results under clean scenario across different fine-grained tasks. Top-2 results are bolded and underlined, respectively.

Model	Discriminative			Generative			Overall
	YNQ	MCQ	Avg	FFQ	IC	Avg	
Otter [24]	0.714	0.682	0.698	0.560	0.336	0.560	0.652
Shikra-7B	0.767	0.594	0.681	0.650	0.460	0.650	0.670
LLaVA-1.5-7B [30]	0.773	0.802	0.788	0.633	0.509	0.633	0.736
InternLM-XComposer-VL-7B	0.872	0.850	0.861	0.606	0.446	0.606	0.776
InstructBLIP-Vicuna-13B [11]	0.730	0.666	0.698	0.733	0.670	0.733	0.710
InstructBLIP-Vicuna-7B [11]	0.704	0.725	0.715	0.733	0.726	0.733	0.721
LLaVA-1.5-13B [30]	0.814	0.861	0.838	0.663	0.576	0.663	0.779
mPLUG-Owl2 [49]	0.811	0.818	0.814	0.706	0.614	0.706	0.778
InternVL2-8B [10]	0.864	0.866	0.865	0.629	0.653	0.629	0.786
InstructBLIP-Flan-T5-XL [11]	0.866	0.845	0.855	0.604	0.753	0.604	0.772
Phi-3-Vision [1]	0.880	0.889	0.884	0.709	0.603	0.709	0.826
InstructBLIP-Flan-T5-XXL [11]	0.882	0.829	0.855	0.719	0.737	0.719	0.810
Qwen-VL-7B [4]	0.867	0.824	0.845	0.767	0.774	0.767	0.819
InternVL3-8B [52]	0.928	0.929	0.928	0.750	0.727	0.750	0.869
InternLM-XComposer2-VL-7B [15]	0.920	0.925	0.922	0.794	0.733	0.794	0.880
InternVL3-14B [52]	0.935	0.947	0.941	0.741	0.750	0.741	0.874
MiniCPM-Llama2-V2.5 [48]	0.921	0.903	0.912	0.744	0.823	0.744	0.856
GLM-4V-9B	0.921	0.928	0.924	0.754	0.804	0.754	0.868
GPT-4o [35]	0.942	0.913	0.928	0.760	0.861	0.760	0.872
DeepSeek-VL2 [44]	0.922	0.919	0.921	0.787	**0.874**	0.787	0.876
Gemini-2.0-flash [12]	0.928	0.942	0.935	0.830	0.871	0.830	0.900
Qwen-VL-Max [2]	**0.943**	**0.951**	**0.947**	**0.846**	0.840	**0.846**	**0.913**

Table 6: Evaluation results under clean scenario across various fine-grained faithful visual perceptual aspects. Top-2 results are bolded and underlined, respectively.

Model	Type	Existence	Color	Shape	Count	Orientation	Posture	Expression	OCR	Spatial Relation	Size Relation	Interaction Relation
Gemini-2.0-flash [12]	0.938	**0.973**	**0.920**	0.885	0.945	0.913	**0.933**	**0.913**	0.955	0.773	0.785	0.850
Qwen-VL-Max [2]	0.925	0.953	0.908	0.893	0.983	**0.935**	0.930	0.750	0.948	**0.833**	**0.800**	0.845
GPT-4o [35]	0.890	0.875	0.915	**0.930**	**0.948**	0.900	0.920	0.843	0.970	0.823	0.748	0.853
DeepSeek-VL2 [44]	0.875	0.948	0.918	0.928	0.903	0.873	0.930	0.825	**0.980**	0.745	0.728	**0.883**
GLM-4V-9B	0.938	0.928	0.913	0.893	0.843	0.830	0.920	0.745	0.930	0.770	0.628	0.873
MiniCPM-Llama2-V2.5 [48]	0.905	0.865	0.890	0.898	0.908	0.763	0.920	0.830	0.890	0.738	0.715	0.838
InternVL3-14B [52]	0.860	0.875	0.875	0.813	0.890	0.898	0.888	0.805	0.913	0.755	0.693	0.860
InternLM-XComposer2-VL-7B [15]	0.890	0.935	0.853	0.853	0.855	0.785	0.873	0.838	0.910	0.755	0.763	0.798
InternVL3-8B [52]	0.873	0.900	0.868	0.833	0.883	0.865	0.900	0.830	0.905	0.750	0.635	0.820
Qwen-VL-7B [4]	**0.963**	0.925	0.885	0.873	0.803	0.683	0.870	0.745	0.945	0.543	0.650	0.788
InstructBLIP-Flan-T5-XXL [11]	0.948	0.923	0.888	0.883	0.860	0.738	0.800	0.845	0.853	0.498	0.638	0.693
Phi-3-Vision [1]	0.830	0.850	0.865	0.813	0.788	0.678	0.830	0.658	0.898	0.710	0.740	0.780
InstructBLIP-Flan-T5-XL [11]	0.950	0.933	0.885	0.865	0.823	0.590	0.798	0.820	0.870	0.468	0.505	0.725
InternVL2-8B [10]	0.808	0.823	0.808	0.715	0.773	0.620	0.830	0.770	0.888	0.608	0.608	0.765
mPLUG-Owl2 [49]	0.905	0.828	0.860	0.803	0.670	0.573	0.780	0.820	0.888	0.412	0.485	0.723
LLaVA-1.5-13B [30]	0.833	0.865	0.815	0.760	0.678	0.560	0.795	0.753	0.855	0.498	0.583	0.713
InstructBLIP-Vicuna-7B [11]	0.908	0.835	0.865	0.793	0.680	0.560	0.833	0.728	0.808	0.400	0.490	0.720
InstructBLIP-Vicuna-13B [11]	0.913	0.888	0.818	0.800	0.540	0.568	0.773	0.720	0.758	0.378	0.560	0.620
InternLM-XComposer-VL-7B	0.815	0.725	0.798	0.798	0.780	0.488	0.770	0.705	0.823	0.535	0.440	0.633
LLaVA-1.5-7B [30]	0.780	0.773	0.780	0.715	0.643	0.503	0.798	0.765	0.783	0.380	0.483	0.685
Shikra-7B [7]	0.763	0.743	0.758	0.598	0.553	0.540	0.753	0.778	0.568	0.375	0.453	0.673
Otter [24]	0.753	0.723	0.670	0.633	0.565	0.443	0.630	0.615	0.630	0.215	0.468	0.523

Table 7: Evaluation results under clean scenario across various fine-grained factual knowledge domains. Top-2 results are bolded and underlined, respectively.

Model	Sports	Politics	Entertainment	Religion	Geography	Culture
Qwen-VL-Max [2]	0.781	**0.875**	**0.916**	**0.984**	**0.961**	0.893
DeepSeek-VL2 [44]	**0.813**	0.750	0.837	0.935	0.928	0.893
GLM-4V-9B [16]	0.656	0.806	0.868	0.910	0.913	0.821
Gemini-2.0-flash [12]	0.781	0.694	0.815	0.978	0.932	0.839
MiniCPM-Llama2-V2.5 [48]	0.781	0.813	0.873	0.957	0.911	0.848
InternLM-XComposer2-VL-7B [15]	0.781	0.785	0.825	**0.984**	0.872	0.839
InternVL3-14B [52]	0.594	0.743	0.789	0.832	0.916	0.848
Qwen-VL-7B [4]	0.750	0.743	0.788	0.861	0.821	0.884
InternVL3-8B [52]	0.594	0.569	0.662	0.769	0.891	0.821
mPLUG-Owl2 [49]	0.688	0.764	0.765	0.967	0.827	0.750
GPT-4o [35]	0.250	0.382	0.620	0.927	0.957	**0.911**
InternVL2-8B [10]	0.625	0.674	0.734	0.804	0.849	0.777
InstructBLIP-Flan-T5-XXL [11]	0.500	0.694	0.705	0.840	0.805	0.768
InstructBLIP-Flan-T5-XL [11]	0.594	0.653	0.721	0.766	0.812	0.786
LLaVA-1.5-13B [30]	0.688	0.625	0.749	0.940	0.774	0.759
InstructBLIP-Vicuna-7B [11]	0.594	0.708	0.763	0.747	0.799	0.750
InstructBLIP-Vicuna-13B [11]	0.406	0.646	0.729	0.644	0.805	0.732
LLaVA-1.5-7B [30]	0.500	0.556	0.591	0.929	0.798	0.750
InternLM-XComposer-VL-7B	0.531	0.660	0.622	0.715	0.748	0.732
Phi-3-Vision [1]	0.219	0.472	0.633	0.522	0.787	0.786
Otter [24]	0.344	0.535	0.492	0.549	0.629	0.625
Shikra-7B [7]	0.406	0.403	0.418	0.742	0.614	0.634